

Experiment 4

Aim: Perform Part of Speech Tagging using NLTK.

Theory:

Part of Speech (POS) Tagging is a key process in Natural Language Processing (NLP) that involves assigning grammatical labels to words in a sentence based on their context and usage. These labels indicate the role of each word, such as noun, verb, adjective, adverb, pronoun, preposition, or conjunction.

The **Natural Language Toolkit (NLTK)** provides built-in tools for POS tagging using statistical and rule-based approaches. POS tagging helps computers understand the syntactic structure of sentences and is essential for many advanced NLP tasks.

Working of POS Tagging in NLTK

NLTK uses pre-trained models to analyze the context of words and assign appropriate POS tags. The commonly used function `pos_tag()` takes a list of word tokens as input and returns pairs of words along with their corresponding POS tags.

Common POS Tags

- **NN / NNS** – Noun (singular / plural)
- **VB / VBD / VBG / VBN / VBP / VBZ** – Verb forms
- **JJ** – Adjective
- **RB** – Adverb
- **PRP** – Pronoun
- **IN** – Preposition

Applications of POS Tagging

- Syntax analysis and parsing
- Information extraction
- Named entity recognition
- Machine translation
- Text summarization

Source Code and Outputs:

```
[1]
✓ 7s
!pip install -U nltk

Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Collecting nltk
  Downloading nltk-3.9.2-py3-none-any.whl.metadata (3.2 kB)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.1)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex<=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)
Downloading nltk-3.9.2-py3-none-any.whl (1.5 MB)
1.5/1.5 MB 10.5 MB/s eta 0:00:00
Installing collected packages: nltk
  Attempting uninstall: nltk
    Found existing installation: nltk 3.9.1
    Uninstalling nltk-3.9.1:
      Successfully uninstalled nltk-3.9.1
  Successfully installed nltk-3.9.2

[2]
✓ 8s
import nltk

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]   /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger.zip.
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]   /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
```

```
[3]
✓ 0s
from nltk.tokenize import word_tokenize
from nltk import pos_tag

text = "NLTK is a powerful Library for natural language processing "

words = word_tokenize(text)
pos_tags = pos_tag(words)

print("POS Tagged Words:")
print(pos_tags)

POS Tagged Words:
[('NLTK', 'NNP'), ('is', 'VBZ'), ('a', 'DT'), ('powerful', 'JJ'), ('Library', 'NNP'), ('for', 'IN'), ('natural', 'JJ'), ('language', 'NN'), ('processin

[4]
✓ 0s
for word,tag in pos_tags:
    print(f"{word}->{tag}")

NLTK->NNP
is->VBZ
a->DT
powerful->JJ
Library->NNP
for->IN
natural->JJ
language->NN
processing->NN
```

Learning Outcomes:

I am able to perform Part of Speech (POS) tagging using NLTK by assigning grammatical labels to each word in a sentence, helping in understanding sentence structure and improving the accuracy of downstream NLP tasks.